

	SUPERCONDUCTING MATERIALS: • Introduction – Properties • Meissner effect • Type I & Type II superconductors • Applications.
1.	Define: (1) Superconductivity (2) Superconductor (3) Residual resistivity Sol: (1) Superconductivity: Phenomena in which certain metals, alloys, or ceramics conduct electricity without resistance when they are cooled below a certain critical temperature is defined as Superconductivity. (Liquid Helium and Liquid Nitrogen are used to achieve very low temperatures) (2) Superconductor : A superconductor is a material that loses all its resistance (offers zero resistance) to the flow of electric current when it is cooled below a certain temperature. Examples: Mercury (Hg), Zinc (Zn), Vanadium (V), Tin (Sn) and Niobium (Nb). (3) Residual resistivity: The electrical resistivity of normal metal decreases steadily as the temperature is decreased and reaches a low value at 0 K called the residual resistivity.
2.	Define: (1) Critical Temperature or Transition temperature T_c. (2) Critical Magnetic field (3) Critical Current density. Sol: • Critical Temperature (T_c): Temperature at which materials electrical resistivity drops down to almost zero is called Critical Temperature. • Critical Magnetic Field (H_c): Minimum magnetic field required to destroy a superconducting state is called critical magnetic field. • Critical Current Density (J_c): Maximum current per unit area flowing through a superconductor without destroying its superconducting state is called Critical Current Density.
3.	Explain any Eight properties of superconductor. Sol: Properties of Superconductors

1. Electrical Resistance

- Very Low and is of the order of 10^{-7} ohm meter.

2. Effect of Impurities

- When impurities are added, superconductivity is not lost but T_c value is lowered.
- When impurities are added, number of charge carriers will remain same, but due to distortion of lattice, mobility decreases. As a result resistivity increases.
- So, it requires to Decrease temperature to maintain superconductivity.

3. Isotope Effect

- The critical or transition temperature T value of a superconductor is found to vary with its isotopic mass. This variation in T with its isotopic mass is called the isotope effect.

The relation between T and the isotopic mass is given by

$$T_c \propto \frac{1}{M^\alpha} \quad \alpha = \text{Isotopic Constant} = 0.5$$
$$T \propto \frac{1}{\sqrt{M}} \text{ OR } T_c \sqrt{M} \text{ is constant.}$$

where M is the isotopic mass,

i.e., the transition temperature is inversely proportional to the square root of the isotopic mass of a single superconductor.

4. Effect of Pressure

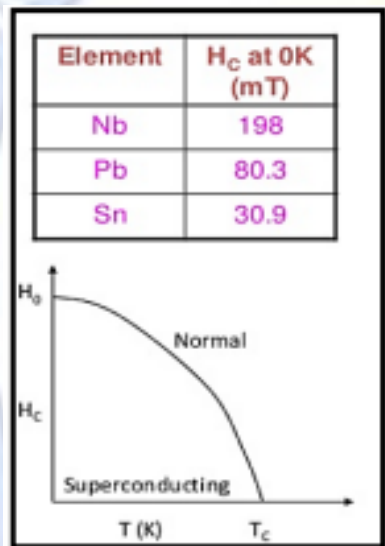
- Certain materials found to exhibit the superconductivity on increasing the pressure over them.
- On increasing pressure, atoms come closer. So, lattice vibrations are reduced. It decreases the resistivity (hence resistance).
- For example, cesium is found to exhibit superconductivity phenomena at $T = 1.5$ K on applying a pressure of 110 Kbar.

5. Magnetic Field Effect

- If a sufficiently strong magnetic field is applied to a superconductor at any temperature below critical temperature, the superconductor undergoes a transition from superconducting state to normal state.
- **Minimum magnetic field required to destroy the superconducting property at any temperature is called critical magnetic field.**

$$H_c = H_{c(0)} \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

- $H_{c(0)}$ = Critical field at 0 K
- T = Temperature
- T_c = transition temperature
- where H_c is the critical field at $T=0K$. The critical field decreases with increasing temperature and becomes zero at $T = T_c$
- The figure shows the variation of critical field H_c as a function of temperature.
- The material is said to be in the superconducting state within the curve and is non-superconducting (i.e., normal state) in the region outside the curve.



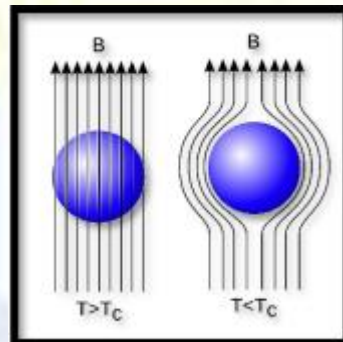
6. Critical Current Density

- It can be defined as **Maximum current per unit area that can be permitted in a superconducting material without destroying its superconducting state.**
- When current density through a superconductor exceeds a critical value J_c , Superconducting state is found to disappear in the sample
- Because current flow generates magnetic field
- At High current density magnetic field will start exceeding the value of H_c
- So, the superconducting state will start disappearing
- Silsbee's Law Derives relation between I_c , J_c and H_c
- For a superconducting wire of radius 'r', Using Ampere's circuital Law

- $H_c = \frac{I_c}{2\pi r}$
- $I_c = 2\pi r H_c$
- $J_c = \frac{I_c}{Area} = \frac{2\pi r H_c}{\pi r^2}$
- $J_c = \frac{2H_c}{r}$

7. Meissner Effect

- **Complete expulsion of all magnetic field lines by a superconducting material is called Meissner effect**
- Magnetic field lines per unit are called magnetic flux



- When a superconducting material is placed in a magnetic field at room temperature, the magnetic field lines penetrate through the material normally.
- But when temperature is lowered to T_c and $H < H_c$, material is found to reject all magnetic field through it.
- When magnetic field is applied to the material, due to passing of surface current, magnetic field is created within the superconducting material
- This magnetic field will counter the applied magnetic field, they cancel each other everywhere inside the material
- **To prove $\chi_m = -1$ (perfect diamagnetism) for superconductors**
- Magnetic induction or magnetic flux density (B) is given by:

$$B = \mu_0(H + M)$$

μ_0 = Permeability of free space

M = Intensity of magnetisation

H = Applied magnetic field

→ For superconductor $B = 0$

$$0 = \mu_0(H + M)$$

But μ_0 can't be zero ($\mu_0 = 4\pi \times 10^{-7}$)

$$\therefore M + H = 0$$

$$\therefore M = -H$$

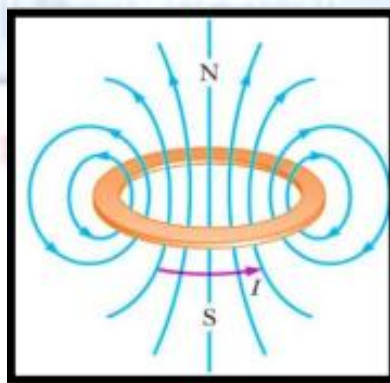
$$\therefore \frac{M}{H} = -1 \quad \left(\chi_m = \frac{M}{H} \right)$$

$$\therefore \chi_m = -1$$

- Hence, $\chi_m = -1$ where $\chi_m = M/H$ is called the magnetic susceptibility.
- Thus, this means that for a superconductor, the susceptibility is negative and maximum, i.e., a superconductor exhibits perfect diamagnetism

8. Persistent Current

- When current is made to flow in superconducting ring which is at temperature equal to or less than T_C , it is observed that current flows through the material without any significant loss in value. **This steady flow of current in a superconducting material is called Persistent current.**
- Current can be induced in a superconductor by applying magnetic field to it and then switching off the magnetic field.
- When the magnetic field is switched off, the field outside the ring disappears but field inside the ring is trapped.
- The collapse of magnetic field outside the ring induces a large current inside the ring itself and maintains the trapped flux.
- Once the current is induced it persists for indefinite period (till its superconductivity state is maintained).



4. Based on what factor superconductors are classified? Explain the types, which one has more applications?

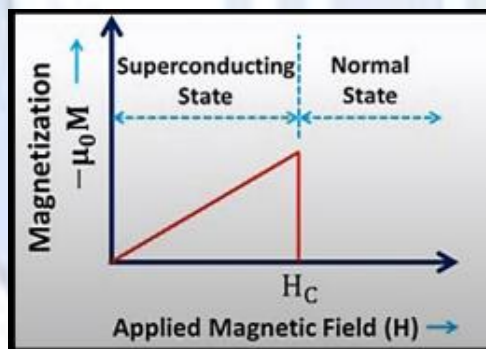
Sol:

Classification of superconductor:

Based on the behaviour of superconducting materials in applied magnetic field, it can classify in two parts.

Type I Superconductor

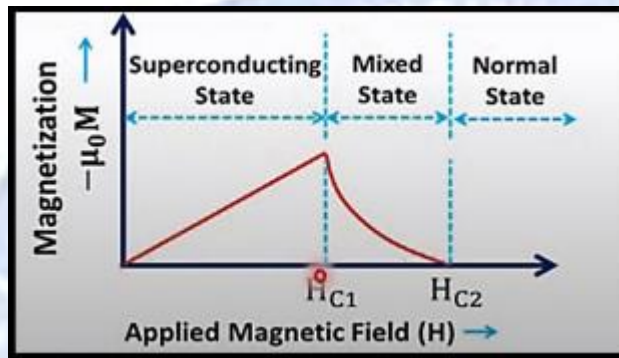
- Also known as soft superconductors
- They exhibit complete Meissner Effect
- Transition from superconducting state to normal state occurs very sharply at critical value
- The Value of H_C is very low (in order of 0.1 T)
- They have Limited applications because of low strength field value
- They allow field penetration easily
- Eg Lead(Pb), Tin(Sn), Vanadium(V)



Type II Superconductor

- They are known as Hard Superconductors
- There are two critical fields H_{C1} (Lower critical field) and H_{C2} (Higher critical field)
- Till H_{C1} material is in superconducting state
- Between H_{C1} and H_{C2} material is in mixed state
- Above H_{C2} material is in normal state
- Meissner effect is partially exhibiting in this kind of material.

- The value of H_{C2} is 100 times or even more as compared to type superconductors.
- They have Wide applications because of high strength field value
- Eg $Y_2Ba_2Cu_3O_7$ (Yttrium Barium Copper Oxide)
- Nb-Ti (Niobium Titanium)



5. Differentiate/Compare: Type I and Type II superconductors.

OR

Compare/Differentiate between Hard and Soft superconductors.

Sol:

<u>Type I Superconductor</u>	<u>Type II Superconductor</u>
Soft Superconductors	Hard Superconductors
One critical field exists	Two critical field exists
Critical field value is very low	Critical field value is very high
They exhibit perfect and complete Meissner Effect	They do not exhibit perfect and complete Meissner Effect
Limited applications because of low strength field value	Wide applications because of high strength field value
Transition occurs Sharply from superconducting state to normal state. Eg. Pb, Hg, Zn, Nb	Transition occurs Smoothly from superconducting state to normal state Eg. $Y_2Ba_2Cu_3O_7$
A graph showing Magnetization ($-\mu_0 M$) on the y-axis versus Applied Magnetic Field (H) on the x-axis. The curve rises linearly through the Superconducting State and drops sharply to zero at H_C in the Normal State. A vertical dashed line marks H_C.	A graph showing Magnetization ($-\mu_0 M$) on the y-axis versus Applied Magnetic Field (H) on the x-axis. The curve rises linearly through the Superconducting State, reaches a peak at H_{C1}, then decreases through the Mixed State, and finally reaches zero at H_{C2} in the Normal State. Vertical dashed lines mark H_{C1} and H_{C2}. Horizontal dashed lines indicate the magnetization levels in each state.

6. Write down the applications of superconductors.

Sol:

- **Power transmission:** Superconductors can be used to transmit power over long distances with little to no loss.
- **Maglev trains:** Superconducting electromagnets are used in magnetic levitation trains (maglev). Maglev trains have less wear and tear and are less likely to derail than other trains.
- **MRI and NMR machines:** Superconductors are used to create strong magnetic fields for magnetic resonance imaging (MRI) and nuclear magnetic resonance (NMR) machines.
- **Particle accelerators:** Superconducting magnets are used in high-energy particle accelerators.
- **Computers:** Superconductors can be used as memory or storage elements in computers.
- **Other applications:** Superconductors are also used in generators, transformers, wireless receivers, inductive heating systems, and more.

7. What is SQUID? Explain with diagram?

OR

Give formation and application of SQUID.

Sol:

Superconducting Quantum Interference Device (SQUID)

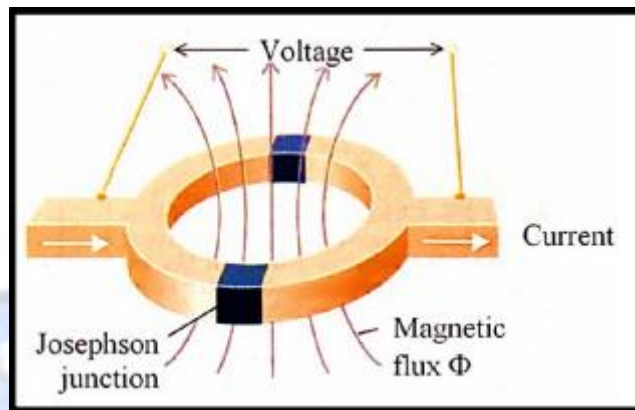
- It works on the principle of Josephson Effect.

Josephson Effect:

- Josephson Effect was defined by Brian Josephson, in the year 1962.
- Superconductors are separated by a thin insulating layer
- Tunnelling of electron pairs takes place from one superconductor to another
- Cooper pairs of electrons tunnel through the insulating layer
- The super-current (J) which is developed across the insulating junction depends on phase difference
- This is called Josephson junction or SIS junction

Construction:

- SQUID is formed when two superconductors and insulators are arranged in such a way that they form two parallel Josephson junction.
- Connected to measuring device which measure voltage changes and magnetic fields across junction



WORKING:

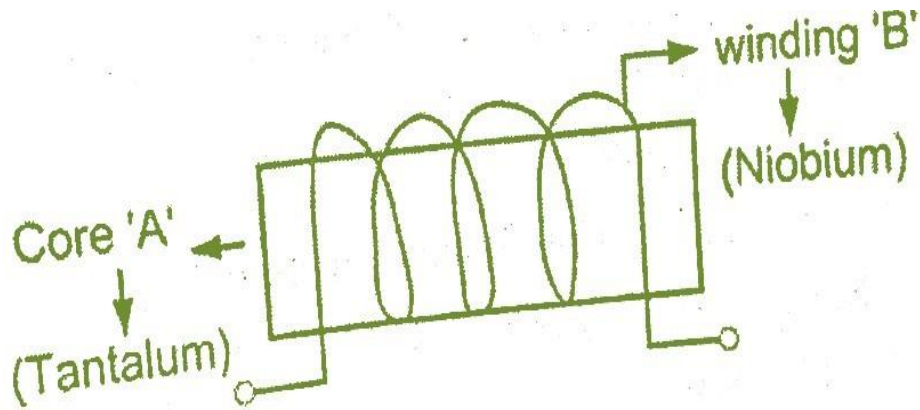
- When current is passed through one side of superconductor, it flows equally through the another side
- Very sensitive to magnetic flux
- Small change in flux will lead to a change in current
- Hence voltage is developed across junction
- Current through the SQUID is highly sensitive to magnetic flux through the closed circuit
- Extremely small magnetic flux can be detected with the device
- As magnetic field, voltage and current are dependent on each other, we can measure any one when other is known
- Detect voltages as small as 10^{-15} V. Magnetic field changes as small as 10^{-21} T can be detected

APPLICATION:

- Used to measure earth's magnetic field.
- It can detect weak magnetic field produced by biological currents like human brain.
- Detection of iron deposition in human liver.

8. Explain Construction and working of Cryotron.

Sol:



- Two different superconducting materials are taken
- A= Tantalum Core and B= Niobium as winding wire.
- Taken them as such that: $H_{cA} < H_{cB}$
- If the current passes through winding B, it induces magnetic field.
- If this induced magnetic field is greater than H_{cA} , its corresponding superconducting state will be destroyed, thereby increasing resistivity and hence decreasing the current and vice-versa.
- Thus we can say that current through the winding controls superconducting state of core.

Uses:

Used in circuit breakers, relays, etc.